

# NEWTON Magnetic Mirror Source Model Technical Documentation

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# 1 Model Purpose and Scope

The NEWTON magnetic mirror source model calculates the fast ion and fusion neutron source produced by neutral beam injection into an axisymmetric magnetic mirror. It connects magnetic geometry, neutral beam attenuation, kinetic fast ion evolution, electrostatic confinement, end losses, expander populations, fusion reactions, and neutron emission in one computational workflow. The primary output is a physically normalized neutron source that can be used by OpenMC for downstream neutronics calculations.

The fast ion model follows the magnetic mirror FBIS framework developed by Egedal et al. for weakly collisional, neutral beam driven mirrors [5]. The broader device context is consistent with the WHAM and BEAM studies, which use high field axisymmetric mirrors, neutral beam heating, ambipolar electrostatic confinement, and expanding end regions [6, 7]. The source model extends this framework with coupled fast deuterium and tritium states, self-consistent beam attenuation, source connected expander populations, arbitrary distribution fusion calculations, and correlated neutron source export.

The calculation begins from either the analytic Egedal mirror field or a ParaTAN linked on axis coil field. Neutral D and/or T beams are propagated through the plasma and attenuated using species resolved atomic stopping rates. Ionization and charge exchange create fast ion birth distributions that are mapped into the invariant coordinates used by the modal FBIS solver. The attenuation framework follows the standard neutral-beam stopping structure used in fusion plasma modeling [3].

The confined kinetic solution evolves the injected ions in speed and magnetic pitch coordinates. It includes Coulomb drag, energy diffusion, pitch scattering, charge exchange removal, magnetic loss, quasineutral electrostatic confinement, and the feedback between the solved density profile and the orbit-averaged scattering basis. Fast D and fast T are solved as distinct populations with a shared magnetic and electrostatic state.

Particles that cross the magnetic loss boundary are reconstructed as directed lost populations and followed into the expanders. The full device population contains confined ions, directed lost ions, material connected expander ions, and source connected reflected ions. Source disconnected local well topology can be identified, but its steady state occupancy is not calculated because the model does not contain a source or collisional closure for that population.

Fusion rates are calculated directly from the local fast D and T velocity distributions with the Bosch and Hale total fusion cross sections [4]. Neutron energies and directions are calculated from the reacting ion distributions using relativistic two-body kinematics following the arbitrary reactant methodology of Eriksson et al. [8]. The correlated event path uses the packaged ENDF/B-VIII.1 center-of-momentum angular distributions for the neutron producing D-D and D-T reactions.

Two neutron representations are retained. A deterministic axial energy spectrum provides a low noise representation for diagnostics and rate conservation. A correlated event bank retains neutron position, laboratory energy, laboratory direction, reaction identity, and sampled reactant state. The event bank can be written as an OpenMC HDF5 file source. Event probabilities are normalized for sampling, while the absolute physical neutron rate remains a separate normalization quantity.

At a high level, the model follows

$$\begin{aligned} & geometry \rightarrow beam\ attenuation \rightarrow fast\ ion\ source \rightarrow kinetic\ closure \\ & \rightarrow full\ device\ populations \rightarrow fusion \rightarrow neutrons \rightarrow OpenMC\ source \end{aligned}$$

The source model is therefore a kinetic source generator rather than a complete plasma or neutronics simulation. It does not solve macroscopic plasma stability, turbulent radial transport, self-consistent finite-beta magnetic equilibrium, or neutron transport. Those effects must be supplied by other models when required by a study.

## 2 Software Architecture and Model Inputs

The source model is organized as a staged physics pipeline rather than as one calculation. Geometry, neutral beam transport, modal kinetics, electrostatic closure, expander populations, fusion, neutron generation, and OpenMC export are implemented in separate packages, while the integration layer controls their ordering and the fixed point iterations that couple them. This structure keeps the physical responsibility of each module clear and makes it possible to trace a final neutron source back to the geometry, beam, kinetic, and electrostatic states that produced it.

The main calculation is started through

```
run_source_model_pipeline()  
source_model_revamp.integration.pipeline
```

It receives a validated `SourceModelRunConfig` and returns a `SourceModelPipelineResult`. The typed stage objects between those two points define where physical state and normalization authority reside. Geometry owns the magnetic grid and cell volumes, the beam stage owns the external fast ion birth source, the operating point stage owns the coupled kinetic and electrostatic state, the fusion stage owns reaction rates, and the neutron stage owns the deterministic and correlated neutron representations.

The high level software flow is

```
YAML input → SourceModelRunConfig → GeometryStageResult  
→ OperatingPointStageResult → FusionStageResult → NeutronStageResult  
→ OpenMCExportStageResult → SourceModelPipelineResult
```

The operating point stage contains most of the nonlinear coupling. At a selected electron temperature it iterates beam attenuation, the modal fast ion solution, density closure, Eq. 42 basis feedback, Eq. 68 current balance, Eq. 70 quasineutrality, ion losses, expander populations, and terminal current until they represent one consistent plasma state. Fusion and neutron stages are downstream consumers of that state and do not redefine its physical normalization.

The YAML configuration therefore contains both physical inputs and numerical controls. Physical inputs define quantities such as the magnetic field, plasma size, neutral beams, startup plasma state, and selected closure options. Numerical inputs define grid sizes, basis resolution, relaxation, convergence tolerances, event statistics, and output behavior. Strict parsing and cross block validation are used so that misspelled, obsolete, or physically incompatible inputs do not silently alter the calculation.

### 2.1 Package responsibilities

The main packages have the following roles.

**geometry/** Magnetic field construction, throat finding, axial domains, flux tube geometry, and plasma facing boundaries

**beam/** Beam geometry, atomic stopping, attenuation, charge exchange bookkeeping, and fast ion birth sources

**orbits/ and scattering/** Magnetic invariants, bounce averages, the  $\Lambda$  to  $\eta$  mapping, and the physical pitch scattering basis

**fbis/ and fbis/modal/** Velocity grids, collision states, modal FBIS, Eq. 59, Eq. 42 feedback, losses, Eq. 70, and local reconstruction

**losses/** Ion and electron loss bookkeeping and loss diagnostics  
**electrostatic/** Current balance and shared electrostatic helpers  
**expander/** source connected expander trajectories, residence mapping, and expander quasineutrality  
**coupling/** Electron energy accounting and the scalar electron temperature solve  
**fusion/** Arbitrary-distribution D-D and D-T fusion-rate integration  
**nuclear\_data/ and neutrons/** Evaluated angular data, neutron kinematics, deterministic spectra, and correlated neutron events  
**export/** Event sidecars and OpenMC HDF5 source writing  
**integration/** Configuration, stage ordering, coupled iterations, validation, qualification, and output assembly

The integration layer is further separated into typed configuration objects, modal stage coupling, and pipeline stages. This keeps lower level physics routines independent from the orchestration required to solve one complete operating point.

## 2.2 Model inputs

The model is configured through a YAML `source_model` block. Strict parsing rejects unknown keys. The principal input groups are

**geometry** Field model,  $B_0$ , throat field, mirror ratio, midplane radius, axial resolution, and ParaTAN-linked geometry controls  
**plasma\_boundaries** Automatic or explicit plasma facing boundary selection  
**plasma\_closure** Startup D and T density state, ion temperature, electron density initial state, and electron temperature initial guess  
**beams** Beam species, power, parent energy, energy components, beamline, radius, divergence, pitch treatment, and atomic data model  
**kinetic\_electrostatic** Modal basis size, velocity grids, Eq. 59 controls, Eq. 42 iteration, Eq. 70 controls, ion loss settings, local reconstruction, current balance, and expander numerics  
**beam\_density\_coupling** Outer fixed point relaxation and convergence tolerances for the solved plasma and beam source  
**power\_balance** Fixed or self-consistent electron temperature closure, external electron heating, and scalar root search controls  
**fusion** Relative gyroangle resolution and pair batching  
**neutrons** Energy bounds, energy resolution, angular model, correlated event count, random seed, gyroangle sampling, and radial source profile  
**openmc\_export** File source output location, filenames, source time, and overwrite behavior  
**output** Reporting and optional iteration-history controls

For beam component  $k$ , the component energy is  $E_k = E_{\text{beam}} f_{E,k}$  and its injected neutral rate is  $\dot{N}_{0,k} = P_k/E_k$ . Component power fractions must sum to one.

For the NBI supported stationary closure, configured D and T density profiles are startup seeds. They initialize the first stopping and collision state but do not become maintained physical populations in the final operating point. Once the kinetic solution exists, the solved quasineutral state becomes the density authority.

Cross block validation also checks conditions such as unique enabled beam identifiers, normalized component power fractions, a throat field above the midplane field, and compatible plasma and kinetic closure selections.

## 2.3 State ownership

Typed result objects are passed between stages rather than collecting all quantities into one mutable global state. `GeometryStageResult` owns geometry and cell volumes. `BeamEnsembleResult` owns the species grouped beam sources. `KineticStageResult` owns modal solutions, local distributions, density, electrostatic, loss, and full device fast ion states. `OperatingPointStageResult` combines the converged beam, kinetic, expander, and electron energy states. Fusion, neutron, and OpenMC export stages then consume that operating point.

This structure is important for debugging because a final neutron discrepancy can be traced backward through neutron, fusion, kinetic, beam, and geometry states without reconstructing intermediate quantities from a final source file.

## 2.4 Primary pipeline states

The most important stage objects also define where normalization authority changes hands.

`GeometryStageResult` Physical cell volumes and magnetic mapping

`BeamEnsembleResult` Deposited birth rates and attenuation audit

`KineticStageResult` Velocity distributions, density, electrostatic state, and losses

`OperatingPointStageResult` Coupled state at one electron temperature

`FusionStageResult` Scalar reaction rates, source profiles, and fusion power

`NeutronStageResult` Deterministic neutron representation and correlated event bank

`OpenMCExportStageResult` Written transport source and measured export evidence

This ownership is deliberate. A downstream stage is not allowed to redefine an upstream physical normalization merely to make two outputs agree. For example, the neutron stage checks its raw neutron rate against the fusion rate rather than rescaling the spectrum. The OpenMC exporter normalizes source probabilities for sampling while retaining the independently calculated physical neutron rate. Similar checks are used for beam deposition, Eq. 61 to Eq. 63 matching, and local distribution reconstruction.

Warm start information is stored separately from the physical result. A warm state can carry the previous beam target density, Eq. 59 modal state, Eq. 42 profile, collision operators, Eq. 70 potential, and wall barrier. Compatibility is checked before reuse. This allows long nested solves to exploit continuation without making convergence depend on hidden state from an earlier operating point.

### 3 Coordinates, Invariants, Grids, and Conventions

The source model uses several coordinate systems because different parts of the calculation are naturally expressed in different variables. Physical geometry is described by the axial coordinate  $z$ , local velocity distributions use speed  $v$  and signed pitch  $\xi = v_{\parallel}/v$ , and the bounce-averaged kinetic calculation uses magnetic constants of motion. Defining these coordinates carefully is important because later stages repeatedly transform the same physical particle population between invariant and local representations.

The central magnetic invariant is  $\Lambda = \mu B_0/E$ , where  $\mu$  is the magnetic moment and  $E$  is the magnetic only kinetic energy. Together with the normalized field  $\tilde{B} = B/B_0$ , it gives  $\xi^2 = 1 - \Lambda\tilde{B}$  and identifies the magnetic trapped-passing boundary at  $\Lambda_M = 1/R_M$ . Egedal's orbit coordinate  $\eta(\Lambda)$  is then constructed from the normalized bounce time. The  $\Lambda$  to  $\eta$  mapping depends only on the magnetic geometry, while the later Eq. 42 density weighting changes the orbit-averaged scattering operator without redefining that magnetic coordinate.

The numerical implementation also separates invariant and local velocity grids. Eq. 59 evolves modal amplitudes on a source aligned speed grid, while electrostatic reconstruction, fusion, and neutron calculations use local  $(v, \xi)$  distributions. Coupled D and T calculations may use different species speed grids while sharing the magnetic geometry, pitch representation, and electrostatic state. This section therefore establishes the coordinate definitions, array layouts, normalization measures, and unit conventions used throughout the rest of the guide.

#### 3.1 Axial and magnetic coordinates

The physical coordinate  $z$  measures distance along the magnetic axis. For the analytic mirror,  $\zeta = z/L$ , where  $L$  is the mirror half length. The midplane is  $\zeta = 0$  and the throats are  $\zeta = \pm 1$ . For ParaTAN-linked geometry, the same normalized coordinate is constructed from the fitted throat locations.

The normalized field and mirror ratio are

$$\tilde{B}(z) = \frac{B(z)}{B_0}, \quad R_M = \frac{B_M}{B_0}. \quad (3.1)$$

Axial geometry is stored at both cell centers and cell edges because transport-like quantities are cell integrated while material boundaries and magnetic events must often coincide with grid edges.

#### 3.2 Velocity and invariant coordinates

The local velocity satisfies  $v^2 = v_{\parallel}^2 + v_{\perp}^2$  and the signed pitch coordinate is  $\xi = v_{\parallel}/v$ . A gyrotropic local distribution is normalized through

$$n = 2\pi \int_0^{\infty} v^2 dv \int_{-1}^1 f(v, \xi) d\xi. \quad (3.2)$$

The magnetic moment and kinetic energy are  $\mu = mv_{\perp}^2/(2B)$  and  $E = mv^2/2$ . Egedal et al. introduce the dimensionless magnetic invariant [5]

$$\Lambda = \frac{\mu B_0}{E}, \quad \xi^2 = 1 - \Lambda\tilde{B}. \quad (3.3)$$

The magnetic trapped-passing boundary is  $\Lambda_M = 1/R_M$ . At a given position, accessibility requires  $\Lambda\tilde{B}(z) \leq 1$ . A trapped orbit turns where  $\tilde{B}(z_b) = 1/\Lambda$ .

For a nonuniform mirror, the local pitch changes along an orbit. The modal calculation therefore uses the Egedal orbit coordinate [5]

$$\eta(\Lambda) = 1 - \frac{\int_0^\Lambda \tilde{\tau}_b(\Lambda') d\Lambda'}{\int_0^1 \tilde{\tau}_b(\Lambda') d\Lambda'}. \quad (3.4)$$

The mapping runs from  $\eta = 1$  at  $\Lambda = 0$  to  $\eta = 0$  at  $\Lambda = 1$ . The magnetic trapped domain ends at  $\eta_{\text{TP}} = \eta(1/R_M)$ . The  $\Lambda$  to  $\eta$  mapping depends only on the magnetic orbit geometry. Eq. 42 density weighting changes the orbit-averaged collision operator, not this coordinate transformation.

### 3.3 Numerical grids and units

The principal grids are `SpeedGrid`, `PitchGrid`, and `LambdaGrid`. The Eq. 59 solve uses a source aligned stretched speed grid so beam component speeds are represented without forcing a uniformly fine high energy domain. Local electrostatic reconstruction can use a separate speed grid.

Common array layouts are  $(n_{\text{mode}}, n_v)$  for modal amplitudes,  $(n_v, n_\eta)$  or  $(n_v, n_\Lambda)$  for invariant distributions, and  $(n_z, n_v, n_\xi)$  for local distributions. Coupled D and T calculations can use different species speed grids while sharing the magnetic geometry and electrostatic state.

Internal energies are generally stored in joules. Configuration temperatures and beam energies are given in keV. Position is in meters, magnetic field in tesla, speed in  $\text{m s}^{-1}$ , density in  $\text{m}^{-3}$ , particle rate in  $\text{s}^{-1}$ , and power in watts. The later electrostatic mapping uses the conserved total ion energy  $U$ , which is kept distinct from the magnetic only kinetic energy  $E$ .

The most common numerical layouts are summarized below.

|                                |                                       |
|--------------------------------|---------------------------------------|
| Modal speed amplitudes         | $(n_{\text{mode}}, n_v)$              |
| Invariant distribution         | $(n_v, n_\eta)$ or $(n_v, n_\Lambda)$ |
| Beam source                    | $(n_z, n_v, n_\Lambda)$               |
| Local invariant distribution   | $(n_z, n_v, n_\Lambda)$               |
| Local pitch distribution       | $(n_z, n_v, n_\xi)$                   |
| Deterministic neutron spectrum | $(n_z, n_E)$                          |

The code includes units in many public field names. Examples include `z_centers_m`, `energies_J`, and `physical_total_rate_s`. This naming convention is part of the interface contract and is useful when tracing quantities across stages with different natural units.

## 4 Magnetic Geometry and Orbit Model

The magnetic geometry defines both the physical volume used for source normalization and the orbit structure used by the kinetic model. The source model supports an analytic Egedal mirror profile and a ParaTAN-linked coil representation. The two field models are constructed differently, but both are reduced to the same normalized central mirror coordinates before FBIS is solved so that the orbit and modal machinery can operate on a common representation.

The geometry stage identifies the magnetic midplane, the left and right throats, the confined kinetic interval, the larger full device interval, the expanding magnetic flux tube, and the axial cell volumes. The confined interval is the domain of the bounce-averaged kinetic and central Eq. 70 calculations. The full device interval extends into the expanders and includes the material and magnetic events needed for downstream trajectory mapping. This distinction allows the central kinetic problem to remain tied to the mirror throats while the final fast ion and neutron sources extend to the physical device boundaries.

Once  $B(z)$  is known, the orbit calculation determines passing and trapped regions in  $\Lambda$ , trapped-particle turning points, bounce times, and orbit measures. These quantities define the  $\Lambda$  to  $\eta$  mapping and the spatial weights used later by the scattering operator. The implementation treats ordinary turning point

singularities and the logarithmic behavior at the exact magnetic separatrix explicitly so that the integrated phase space measure remains numerically well defined.

#### 4.1 Analytic and ParaTAN-linked fields

The analytic benchmark uses the Egedal Eq. 56 profile [5]

$$\tilde{B}(\zeta) = 1 + (R_M - 1) \frac{\exp[-(1 - |\zeta|)^k/w^k] - \exp[-1/w^k]}{1 - \exp[-1/w^k]}, \quad |\zeta| \leq 1. \quad (4.1)$$

It satisfies  $\tilde{B}(0) = 1$  and  $\tilde{B}(\pm 1) = R_M$ . The parameters  $w$  and  $k$  control the approach from a square well to a smoother physical mirror.

The ParaTAN-linked option derives effective coil locations and radii from the surrounding device geometry. Each effective circular coil contributes the standard on axis field

$$B_z(z) = \frac{\mu_0 I_{\text{eff}} R^2}{2 [R^2 + (z - z_c)^2]^{3/2}}. \quad (4.2)$$

Relative coil amplitudes are first adjusted to reproduce the requested mirror ratio, then a common scale is applied to match the requested midplane field. Left and right throat maxima are found and refined independently. The central well is required to rise monotonically from the midplane toward each throat and the two fitted mirror ratios must satisfy the configured symmetry tolerance.

For fitted throat locations  $z_L$  and  $z_R$ , the confined coordinate is defined by  $z_c = (z_R + z_L)/2$ ,  $L = (z_R - z_L)/2$ , and  $\zeta = (z - z_c)/L$ . This maps the physical confined interval back to approximately  $-1 \leq \zeta \leq 1$ .

#### 4.2 Confined and full device geometry

The confined grid spans the two magnetic throats and is used by the modal kinetic and central electrostatic calculations. The full device grid can extend into both expanders. Important magnetic events, plasma facing boundaries, material intersections, and device transitions are inserted as exact grid edges before remaining intervals are subdivided.

The reduced axisymmetric plasma column assumes magnetic flux conservation. If the midplane cross sectional area is  $A_0$ , then

$$A(z) = \frac{A_0}{\tilde{B}(z)}, \quad a(z) = \frac{a_0}{\sqrt{\tilde{B}(z)}}. \quad (4.3)$$

Axial cell volumes are  $\Delta V_i = A(z_i) \Delta z_i$ . The same volumes are used by density normalization, beam source averaging, particle inventories, fusion rates, and neutron rates.

#### 4.3 Bounce motion and numerical orbit treatment

For a passing orbit with  $\Lambda \leq 1/R_M$ , the positive integration limit is the right throat. For a trapped orbit,  $\Lambda > 1/R_M$  and the turning point is obtained from  $\tilde{B}(\zeta_b) = 1/\Lambda$ .

The normalized magnetic bounce time is [5]

$$\tilde{\tau}_b(\Lambda) = \int_0^{\zeta_b} \frac{d\zeta}{\sqrt{1 - \Lambda \tilde{B}(\zeta)}}. \quad (4.4)$$

The physical full bounce period is  $\tau_b = 4L\tilde{\tau}_b/v$ . The orbit calculation is magnetic only. The ambipolar potential is introduced later during electrostatic accessibility and local reconstruction.

At an ordinary trapped turning point the integrand has an integrable inverse-square-root singularity. The implementation removes the endpoint stiffness with the mapping  $\zeta = \zeta_b(2u - u^2)$  and evaluates the transformed integral with composite Gauss-Legendre quadrature. The exact magnetic separatrix has a logarithmically divergent point bounce time for a smooth throat maximum. When that separatrix lies on the numerical  $\Lambda$  grid, the code uses a finite cell-averaged value so the integrated phase space measure used by Eq. (3.4) remains well defined.

The orbit utilities also evaluate magnetic orbit averages needed by the later collision operator. Purely magnetic averages use the orbit measure directly. Density weighting is introduced in Section 6 when the Eq. 42 profile is applied. Keeping these operations separate prevents density iteration from altering the magnetic orbit map.

The ParaTAN-linked path retains both physical throat coordinates and the normalized  $\zeta$  representation. This is important because the modal problem assumes the normalized confined interval, while expander and material calculations require the actual device coordinates. The geometry stage therefore acts as the common source of truth for field values, throat locations, axial volumes, and boundary placement used by every later calculation.

## 5 Neutral Beam Attenuation and Fast Ion Source Construction

Neutral beam injection provides the external source that drives the fast ion kinetic model. The beam stage therefore converts a specified set of D and T neutral beams into a spatially and velocity resolved fast ion birth distribution. Each beam is propagated through the physical plasma geometry, including finite beam radius, divergence, multiple energy components, and partial overlap with the expanding flux tube. The resulting path length through each plasma cell determines where the neutral population can interact.

Neutral attenuation is calculated from local ionization, charge-exchange, and other removal rates. Before a solved kinetic target exists, the configured startup plasma supplies the first stopping state. Once local fast D and T distributions are available, heavy particle interaction rates can be recalculated from those represented target distributions. The beam source and plasma state are therefore coupled in the final operating point solve rather than being treated as independent inputs.

The stage also distinguishes fast ion birth from net fueling. Ionization creates a fast projectile ion and adds net plasma inventory. Primary charge exchange creates a fast projectile ion while removing a target ion, so it changes the velocity space population without contributing the same net fueling. These channel resolved rates are mapped into the FBIS source  $S(z, v, \Lambda)$ , while target charge exchange can also become a speed resolved kinetic sink. The discrete source is normalized with the physical velocity space measure so the represented birth rate is checked against the attenuation calculation rather than repaired later by rescaling.

### 5.1 Beam geometry and energy components

A beamline is represented as a finite straight segment  $\mathbf{x}(s) = \mathbf{x}_0 + \hat{\mathbf{u}}s$ . Its intersection with every axial plasma cell is calculated. A finite beam radius and divergence are included, with  $r_b(s) = r_{b,0} + s \tan \theta_{\text{div}}$ . The overlap of the circular beam footprint with the local circular plasma cross section is integrated along the intersected path. The resulting effective attenuation length is

$$\Delta s_{\text{eff}} = \int_{\text{cell}} f_{\text{overlap}}(s) ds. \quad (5.1)$$

Each energy component has  $E_k = E_{\text{beam}} f_{E,k}$  and  $P_k = P_{\text{beam}} f_{P,k}$ . The component rate is  $\dot{N}_{0,k} = P_k / E_k$ . Components are attenuated separately because their velocities and atomic rates differ.

The injection pitch is derived from the beamline or supplied explicitly. The source can use one representative pitch or a finite pitch distribution. The finite width model combines beam divergence and finite

radius angular spread, then applies Gauss-Legendre quadrature over the resulting pitch interval.

## 5.2 Atomic stopping and attenuation

Neutral beam attenuation in a plasma is governed by ionization and charge transfer processes [3]. The implementation groups primary neutral removal into

$$\kappa_{\text{tot}} = \kappa_{\text{ion}} + \kappa_{\text{cx}} + \kappa_{\text{other}}. \quad (5.2)$$

For a reaction-rate coefficient  $K = \langle \sigma g \rangle$ , the corresponding line attenuation coefficient is  $\kappa = nK/v_b$ . Electron impact ionization uses a Maxwellian electron average. Before a solved kinetic target is available, heavy particle rates use the configured startup reference. Once the coupled kinetic state exists, heavy particle ionization and charge exchange can be evaluated from the solved local D and T distributions.

Within one path cell the primary neutral rate obeys  $d\dot{N}_0/ds = -\kappa_{\text{tot}}\dot{N}_0$ . The exact transmitted rate for cell constant coefficients is

$$\dot{N}_{\text{out}} = \dot{N}_{\text{in}} \exp(-\kappa_{\text{tot}} \Delta s_{\text{eff}}). \quad (5.3)$$

The removed fraction is partitioned among physical channels according to  $\kappa_c/\kappa_{\text{tot}}$ . Cells are processed in physical beam traversal order.

The current hydrogenic D and T model resolves electron, D, and T ionization contributions and D and T charge-exchange contributions.

The attenuation stage keeps the physical channel rates separate even when several channels have the same effect on the primary neutral beam. This is required because ionization, projectile charge exchange, target removal, and unrepresented neutral loss have different consequences for the kinetic particle balance. The beam result therefore retains both the total removal rate and its channel decomposition for every beam component and axial cell.

When a solved kinetic target is available, the heavy particle rate calculation integrates the atomic cross section against the represented local target distribution rather than replacing that target with one effective temperature. The resulting rate coefficients are then returned to the same attenuation equation. This closes the feedback between the solved fast ion population and the next beam evaluation without changing the beam transport model itself.

## 5.3 Fast birth, fueling, and FBIS source mapping

Ionization and primary neutral charge exchange both create a fast projectile ion. Their fast ion birth density is

$$q_{\text{fast}} = q_{\text{ion}} + q_{\text{cx}}. \quad (5.4)$$

Net fueling is different. Ionization adds a net ion, while charge exchange replaces a target ion with the fast projectile ion. The net fueling density is therefore  $q_{\text{fuel}} = q_{\text{ion}}$ . When the target is kinetic, the charge-exchange event rate is also mapped into a speed resolved target ion sink so the reconstructed removal rate matches the attenuation event rate.

For beam energy  $E_b$ , the birth speed is  $v_b = \sqrt{2E_b/m_b}$ . At axial location  $z$ , injection pitch  $\theta_b$  maps to

$$\Lambda_b(z) = \frac{\sin^2 \theta_b}{\tilde{B}(z)}. \quad (5.5)$$

Births are deposited onto the discrete speed and  $\Lambda$  grids to form  $S(z, v, \Lambda)$  with shape  $(n_z, n_v, n_\Lambda)$ . The discrete source uses the physical velocity space cell measure so its integral reproduces the attenuation birth rate directly. No later global rescaling is used to repair source deposition.

## 6 Modal FBIS Formulation and Eq. 42 Basis Construction

The modal FBIS formulation reduces the pitch scattering part of the kinetic problem to a finite set of physical eigenmodes. Instead of evolving the full distribution directly on a multidimensional pitch grid, the invariant distribution is written as  $f(v, \eta) = \sum_j f_j(v) I_j(\eta)$ . Each eigenfunction  $I_j$  describes one pitch-space structure, while the amplitude  $f_j(v)$  describes how that structure is populated in speed. The following Eq. 59 section then solves those speed amplitudes.

The basis construction begins from the square mirror Lorentz eigenproblem, whose even reference modes satisfy the loss-cone boundary condition and are indexed by noninteger orders. Those reference functions are mapped onto the physical  $\eta$  interval and used in a conservative weak Galerkin projection of the orbit-averaged Lorentz operator. The physical eigenvalues and eigenfunctions therefore depend on the actual magnetic geometry rather than on the square well approximation alone.

A second dependence enters through Egedal Eq. 42. Because the local scattering frequency scales with plasma density, the orbit average contains the normalized axial density profile. The coefficients  $A(\Lambda)$  and  $D(\Lambda)$  are therefore rebuilt when the solved quasineutral density changes. In the self-consistent closure, the model iterates the density profile, the physical scattering basis, and the kinetic electrostatic state until both the density weighting and the retained eigenbasis converge. The final part of the section projects the physical beam births onto those retained modes and checks that the represented modal source reproduces the confined birth rate without posterior renormalization.

### 6.1 Modal representation

The invariant distribution is expanded as

$$f(v, \eta) = \sum_j f_j(v) I_j(\eta). \quad (6.1)$$

### 6.2 square mirror reference eigenbasis

The physical nonuniform mirror basis is constructed from the square mirror Lorentz problem used in the Egedal formulation [5]. The reference operator is

$$\mathcal{L} = \frac{\partial}{\partial \xi} \left[ (1 - \xi^2) \frac{\partial}{\partial \xi} \right]. \quad (6.2)$$

For the even part of the distribution, the reference mode  $M_\ell(\xi)$  satisfies

$$M_\ell(0) = 1, \quad \left. \frac{dM_\ell}{d\xi} \right|_{\xi=0} = 0, \quad M_\ell(\xi_{\text{TP}}) = 0, \quad (6.3)$$

where

$$\xi_{\text{TP}} = \sqrt{1 - \frac{1}{R_M}}. \quad (6.4)$$

The even reference solution is

$$M_\ell(\xi) = {}_2F_1\left(-\frac{\ell}{2}, \frac{\ell+1}{2}, \frac{1}{2}, \xi^2\right), \quad \lambda = \ell(\ell+1). \quad (6.5)$$

The required orders  $\ell$  are generally noninteger. The implementation determines them numerically from the roots of  $M_\ell(\xi_{\text{TP}}) = 0$ . These square mirror functions are not used as the final pitch eigenfunctions. They provide the reference expansion basis from which the physical nonuniform mirror modes are constructed. The corresponding implementation is in `scattering/square_mirror_basis.py`.

### 6.3 Orbit-averaged Lorentz operator and Eq. 42 coefficients

In the invariant coordinate  $\Lambda$ , the local operator can be written

$$\mathcal{L} = \left(\frac{4}{\tilde{B}} - 6\Lambda\right) \frac{\partial}{\partial \Lambda} + 4\Lambda \left(\frac{1}{\tilde{B}} - \Lambda\right) \frac{\partial^2}{\partial \Lambda^2}. \quad (6.6)$$

Following Egedal Eq. 42, the collision operator is averaged along the magnetic orbit with density weighting [5]. For quantity  $Q$ ,

$$\langle Q \rangle_{z,n} = \frac{1}{\tilde{\tau}_b} \int_0^{\zeta_b} \frac{Q(\zeta) [n(\zeta)/\langle n \rangle]}{\sqrt{1 - \Lambda \tilde{B}(\zeta)}} d\zeta. \quad (6.7)$$

Applying this average to the local operator gives the explicit coefficients

$$A(\Lambda) = 4 \left\langle \frac{1}{\tilde{B}} \right\rangle_{z,n} - 6\Lambda \langle 1 \rangle_{z,n}, \quad (6.8)$$

and

$$D(\Lambda) = 4\Lambda \left[ \left\langle \frac{1}{\tilde{B}} \right\rangle_{z,n} - \Lambda \langle 1 \rangle_{z,n} \right]. \quad (6.9)$$

The factor  $\langle 1 \rangle_{z,n}$  is retained explicitly. With the Eq. 42 definition, the orbit average is normalized by the magnetic bounce time rather than by a second density weighted denominator. It is therefore not generally equal to one for a nonuniform density profile. For uniform density weighting,  $n(z)/\langle n \rangle = 1$  and  $\langle 1 \rangle_{z,n} = 1$ , recovering the magnetic only orbit-averaged coefficients. The coefficient construction is implemented in `scattering/lorentz_operator.py`.

The density scale is normalized by the physical flux tube volumes. The profile is required to be compatible with the symmetric central mirror and is even-averaged before basis construction. For uniform weighting, Eq. (6.7) reduces to the magnetic orbit average.

### 6.4 Physical eigenbasis

The magnetic  $\Lambda$  to  $\eta$  mapping is constructed before density weighting. The physical eigenfunctions then satisfy

$$\langle \mathcal{L} \rangle_z I_j(\eta) = -\lambda_j I_j(\eta), \quad 0 \leq \eta \leq \eta_{\text{TP}}. \quad (6.10)$$

The implementation expands each physical mode in the mapped square mirror reference basis and evaluates a conservative weak Galerkin projection. Its scattering and mass matrices have the forms

$$K_{lm} = - \int_0^{\eta_{\text{TP}}} D \frac{dM_l}{d\Lambda} \frac{dM_m}{d\Lambda} d\eta, \quad G_{lm} = \int_0^{\eta_{\text{TP}}} M_l M_m d\eta. \quad (6.11)$$

The physical modes are then obtained from

$$K\mathbf{V}_j = -\lambda_j G\mathbf{V}_j. \quad (6.12)$$

The retained modes are ordered by increasing positive eigenvalue and normalized with  $I_j(0) = 1$ . Their mapped reference functions vanish at  $\eta_{TP}$ , which enforces the fixed magnetic trapped-passing boundary in the modal representation.

The physical basis is independently qualified by grid refinement, eigenfunction overlap, eigenpair residuals, matrix symmetry, Gram-matrix conditioning, normalization, boundary values, and retained mode convergence. Modes between successive grids are matched by their normalized overlap rather than by assuming that discrete eigenvalue ordering is unchanged. These checks establish that the pitch basis is numerically resolved before it is used by the velocity solver.

The basis result therefore carries more than the final eigenvalues. It also retains the  $\Lambda$  to  $\eta$  map, mode values on the physical interval, boundary slopes used by the ion loss calculation, the Eq. 42 density state, and basis convergence evidence. This shared object is reused across nearby operating point evaluations when its physical definition remains compatible.

## 6.5 Self-consistent Eq. 42 feedback and source projection

For self-consistent density weighting, the first basis is built with uniform density. The kinetic and electrostatic state is solved, its quasineutral positive-charge density is converted into a new normalized Eq. 42 profile, and the basis is rebuilt. The density shape and positive density scale are relaxed separately until both the profile and density-dependent basis satisfy their convergence criteria.

For the NBI supported stationary closure, this Eq. 42 density is built from the solved kinetic D plus kinetic T positive charge. The startup seed is not retained as an added physical density contribution.

The beam source is mapped from  $\Lambda$  to  $\eta$  and projected onto the retained physical modes. Births outside the trapped modal interval are classified as prompt magnetic losses. The projection uses the discrete Gram matrix rather than assuming exact numerical orthogonality. The represented modal source rate is calculated from the retained expansion and compared with the physical confined birth rate as a convergence check rather than being normalized afterward.

## 7 Eq. 59 Velocity Solve and Collision Treatment

Once the physical pitch basis and modal source coefficients are known, Eq. 59 determines the speed dependence of every retained mode. This is the main collisional slowing down calculation. It combines beam injection, electron slowing, ion drag, energy diffusion, pitch scattering loss, and the speed resolved charge-exchange sink. Because the fast ion collision coefficients depend on the distribution being solved, the Eq. 59 problem is nonlinear even though each modal solve is linear when the collision state is fixed.

The current implementation builds collision coefficients from ordered test field pairs. Electrons supply the analytic Spitzer slowing normalization. The solved fast population of the same species supplies the non-Maxwellian self Rosenbluth state. In coupled D and T operation, the solved distribution of the other fast species supplies a reduced isotropic cross species Rosenbluth state. The active stationary ion operator therefore comes from the solved fast populations rather than from a retained thermal reservoir.

The velocity equation is discretized in conservative speed flux form on a source aligned finite volume grid and solved mode by mode with a tridiagonal system. Nonlinear fixed point convergence, speed-grid refinement, upper domain extension, and high speed tail coverage are assessed separately. The same discrete operator is also used to calculate pair resolved energy transfer, including fast ion heating of electrons, so the later electron temperature closure uses energy moments that are consistent with the kinetic equation itself.

## 7.1 Hot ion velocity equation

The reference Eq. 59 form is

$$-\beta_m \lambda_j \frac{v_c^3 \tilde{g}_1}{v^3} f_j + \frac{1}{v^2} \frac{\partial}{\partial v} \left[ (v^3 + v_c^3 \tilde{h}) f_j + \frac{v_c^3 \tilde{g}_2}{2} \frac{\partial f_j}{\partial v} \right] = -\tau_s S_j \frac{\delta(v - v_0)}{v^2}. \quad (7.1)$$

Here  $\tau_s$  is the electron slowing down time,  $v_c$  is the critical velocity, and  $\lambda_j$  is the pitch scattering eigenvalue. The density normalized Rosenbluth quantities are  $\tilde{h} = -v^2 \partial h / \partial v$ ,  $\tilde{g}_1 = \partial g / \partial v$ , and  $\tilde{g}_2 = v^2 \partial^2 g / \partial v^2$ .

In the Egedal construction, the first modal amplitude determines the hot ion Rosenbluth state. Eq. (7.1) is solved, the first mode is updated, and the collision coefficients are recalculated. The resulting problem is nonlinear even though each solve at fixed collision coefficients is linear.

## 7.2 Collision composition in the current implementation

The code builds an ordered collision state for each fast test species. Electrons provide the analytic Spitzer slowing normalization. Ion drag, energy diffusion, and pitch scattering are assembled from explicit pair contributions.

For each active ion field pair  $p$ , the implementation first forms the Rosenbluth weight scales

$$C_{g,p} = \tau_s \Gamma W_{g,p}, \quad C_{A,p} = \tau_s \Gamma (W_{h,p} - W_{g,p}), \quad (7.2)$$

where  $W_{g,p}$  and  $W_{h,p}$  contain the field density, charge ratio, Coulomb logarithm, and test-to-field mass dependence defined by the pair state. The common test species coefficient is

$$\Gamma = \frac{Z_t^4 e^4}{4\pi \epsilon_0^2 m_t^2}. \quad (7.3)$$

The resulting ordered pair contributions are

$$A_p(v) = C_{A,p} \tilde{h}_p(v), \quad D_p(v) = \frac{1}{2} C_{g,p} \tilde{g}_{2,p}(v), \quad P_p(v) = \frac{1}{2} C_{g,p} \tilde{g}_{1,p}(v). \quad (7.4)$$

The code also checks the pair mass identity

$$\frac{C_{A,p}}{C_{g,p}} = \frac{m_t}{m_{f,p}}, \quad (7.5)$$

where  $m_t$  is the fast test species mass and  $m_{f,p}$  is the field species mass. The total ion contributions used by one fast test species are then

$$A(v) = \sum_p A_p(v), \quad D(v) = \sum_p D_p(v), \quad P(v) = \sum_p P_p(v). \quad (7.6)$$

The normalized Rosenbluth functions provide the speed dependence, while the pair weights provide the physical density, charge, mass, and Coulomb log scaling. This ordered pair construction allows the same Eq. 59 operator assembly to represent a Maxwellian reference field, a solved fast self field, or the reduced solved fast field of the other isotope.

For the NBI supported stationary closure, the configured startup D and T populations are not maintained thermal collision populations. The generic pair machinery contains Maxwellian reference D and T records for common bookkeeping, but their active densities are set to zero in this closure. If a nonzero Maxwellian reference field is used by another supported path, its thermal speed and normalized speed coordinate are

$$v_t = \sqrt{\frac{2T}{m}}, \quad x = \left(\frac{v}{v_t}\right)^2, \quad (7.7)$$

and the corresponding  $\tilde{h}$ ,  $\tilde{g}_1$ , and  $\tilde{g}_2$  Rosenbluth functions are evaluated analytically, with small- $x$  series used near the origin to avoid cancellation. In the active NBI supported D and T stationary state, these Maxwellian reference pairs contribute no maintained ion density or collision force.

The active fast self contribution is built from the solved non-Maxwellian distribution. Its Rosenbluth speed dependence is calculated from the current first modal amplitude  $f_1(v)$ , while the physical density that scales the self collision contribution is obtained from the complete retained modal distribution. The first mode therefore controls the normalized isotropic Rosenbluth shape without being treated as the complete fast ion inventory.

When both fast D and fast T are active, the other solved fast species supplies a reduced external field. For a field species first mode distribution  $f_F(v)$ , the normalization used by that reduced state is

$$n_F = 4\pi \int_0^\infty v^2 f_F(v) dv. \quad (7.8)$$

Velocity moments of this field distribution are used to evaluate density normalized isotropic Rosenbluth coefficients on the test species speed grid. D acting on T and T acting on D are separate ordered interactions because the test and field masses enter differently. The external fast field is rebuilt as the other species changes during the coupled iteration. This is a reduced isotropic cross species treatment and does not claim a full anisotropic multispecies Landau operator. The scalar screening and representative Coulomb log quantities use the current solved kinetic species densities. The collision state construction is implemented primarily in `fbis/pairwise_collisions.py`, `fbis/collision_parameters.py`, and `fbis/modal/eq59/`.

The fast self Rosenbluth shape and the physical fast ion density therefore have deliberately separate roles, and the same distinction is retained for the reduced fast cross species field.

With the pair coefficients collected into drag  $A(v)$ , energy diffusion  $D(v)$ , and pitch scattering  $P(v)$ , the implemented mode equation can be viewed schematically as

$$-\lambda_j \frac{P(v)}{v^3} f_j + \frac{1}{v^2} \frac{\partial}{\partial v} \left[ (v^3 + A) f_j + D \frac{\partial f_j}{\partial v} \right] - \tau_s \nu_{cx}(v) f_j = -\frac{\tau_s S_j(v)}{v^2}. \quad (7.9)$$

The charge-exchange frequency  $\nu_{cx}(v)$  is an additional target removal sink constructed by the beam stage. It is not part of the original Coulomb Eq. 59 operator.

### 7.3 Discrete speed solve and nonlinear iteration

The speed equation is discretized on a cell centered, source aligned grid. At fixed collision coefficients, the speed-space flux has the form

$$F_v = b(v) f_j + D(v) \frac{\partial f_j}{\partial v}, \quad b(v) = v^3 + A(v). \quad (7.10)$$

Internal face fluxes use Scharfetter-Gummel fitting so drift and diffusion remain stable across different regimes. Lower and upper speed-face fluxes are zero. Pitch scattering and charge exchange appear as local sinks, which leaves a tridiagonal equation for each retained mode.

If no compatible warm state is available, the fast self Rosenbluth shape is initialized from a Maxwellian reference. Each nonlinear iteration rebuilds the Rosenbluth state and collision operator, solves the linear modal equations, and relaxes the new modal amplitudes toward the previous state. Convergence requires agreement in the modal state, nonlinear residual, fast ion inventory, effective energy, and collision coefficients. Sustained stagnation, oscillation, and divergence are tracked separately from simply reaching the iteration limit.

Warm starts can be reused between nearby density or temperature evaluations when their species, source, basis, eigenvalues, collision representation, and speed grids remain compatible. They reduce runtime but never bypass convergence checks. When a compatible state is moved to a larger speed domain, cell averages are conservatively transferred using the spherical velocity shell measure so the warm start does not create or destroy represented inventory.

The nonlinear state also records the difference between iteration convergence and final numerical qualification. A state can be useful as a continuation seed before all refinement studies have been run. Final qualification requires the designated assessment solve on the selected speed grid and the associated resolution and domain checks.

## 7.4 Velocity grid and energy transfer qualification

A converged nonlinear fixed point on one speed grid is not sufficient. The code separately assesses speed resolution, upper domain extension, high speed particle fraction, and high speed energy fraction. It compares quantities such as particle inventory, effective ion energy, modal loss rate, modal loss power, electron heating, and source-to-loss balance across refined grids.

The same discrete operator is used to evaluate pair resolved energy transfer. The code reconstructs the speed-space flux from each ordered collision contribution and integrates its energy moment. It separately retains fast ion to electron transfer, ion pair exchange, pitch loss power, charge-exchange removal power, and represented source power. A discrete energy identity checks that these terms are consistent with the same operator that produced the Eq. 59 distribution.

This provides the electron heating term used later in the electron energy balance without introducing an independent approximate heating model.

## 8 Density Closure, Ion Losses, and Electrostatic Closure

The Eq. 59 solution must be closed with the density, end loss, and ambipolar electrostatic state. These quantities are mutually dependent. Pitch scattering determines the ion flux across the magnetic loss boundary, that ion flux determines the electron current required for ambipolarity, the electron barrier determines the confined electron density profile, and the electrostatic potential changes the local accessibility of the ion distribution. The source model therefore solves these pieces as one coupled closure rather than as independent corrections.

The ion loss chain follows Egedal Eqs. 61 through 64 [5]. The modal distribution provides the derivative at the fixed magnetic boundary  $\Lambda_M = 1/R_M$ . Eq. 61 converts that derivative into the one end differential loss spectrum with the physical mirror geometry factor  $G(z)$ . Eq. 63 then reconstructs the directed lost ion distribution, and Eq. 64 fixes its amplitude by matching the reconstructed throat flux to the Eq. 61 loss rate in each invariant speed cell. The current implementation also determines the left and right parallel loss temperatures from the represented throat state.

Electron confinement is then obtained from current balance and Eq. 70 quasineutrality. The exact tail refilling electron loss kernel is used to determine the wall barrier, while the Egedal Eq. 68 and Eq. 69 expressions are retained as asymptotic references. Eqs. 71 and 72 modify the locally accessible ion region and the mapping used during local reconstruction, but they do not replace the active fixed magnetic Eq. 61 loss boundary. The final state contains the potential profile, electron density, directed lost populations, local D and T distributions, and the exact midplane and throat anchor states required by the larger operating point iteration.

## 8.1 Density state and fixed magnetic ion loss boundary

The code distinguishes electron midplane density, electron collision density, parent Maxwellian density, local electron density, and species resolved fast ion density. These roles are kept separate because they enter different parts of the kinetic and electrostatic calculation.

The active central ion loss boundary remains the magnetic boundary

$$\Lambda_M = \frac{1}{R_M}. \quad (8.1)$$

Following Egedal Eq. 61, the one end differential ion loss rate is determined by the pitch-space flux through this boundary [5]:

$$\frac{dN_{\text{loss},i}}{dt dv} = 4\pi A_0 \tilde{\Gamma}_\xi l \frac{B_0}{B_M} G(l) v \left. \frac{\partial f}{\partial \Lambda} \right|_{\Lambda_M}. \quad (8.2)$$

Here  $A_0$  is the midplane plasma area and  $\tilde{\Gamma}_\xi$  contains the pitch scattering strength. The magnetic geometry factor from Egedal Eq. 62 is

$$G(z) = \frac{1}{l} \int_{-l}^z \frac{B_0}{B(z')} \sqrt{1 - \frac{B(z')}{B_M}} dz'. \quad (8.3)$$

For the modal distribution, the required boundary derivative is

$$\left. \frac{\partial f}{\partial \Lambda} \right|_{\Lambda_M} = \sum_j f_j(v) \left. \frac{\partial I_j}{\partial \Lambda} \right|_{\Lambda_M}. \quad (8.4)$$

Eq. (8.2) is the bridge between the invariant modal solution and the physical one end loss spectrum. The active boundary slopes come from the conservative weak form eigenbasis so the reconstructed boundary flux remains consistent with the pitch scattering sink represented by the modal eigenvalues. A direct numerical boundary derivative is retained only as an independent diagnostic. The Eq. 61 spectrum is one end loss. A symmetric central mirror therefore has two corresponding loss surfaces. The active reconstruction is implemented in `fbis/modal/solver/solver_loss_rates.py`.

## 8.2 Directed Eq. 63 lost ion population

A boundary crossing rate does not by itself define the downstream velocity distribution. Egedal Eq. 63 introduces the directed lost ion branch [5]

$$f_L^+(U, \Lambda, z) = H(U) \exp \left[ -\frac{U(1 - \Lambda R_M)}{T_L} \right] G(z), \quad \Lambda \leq \Lambda_M. \quad (8.5)$$

The current implementation calculates the left and right parallel loss temperatures  $T_L$  from the represented throat state instead of prescribing them independently. The amplitude  $H(U)$  is then obtained from the Eq. 64 flux-matching condition. Matching is performed in each invariant speed cell so the reconstructed Eq. 63 rate reproduces the corresponding Eq. 61 rate without a posterior normalization.

The total ion energy  $U$  includes the electrostatic energy change. This retains the energy gain of escaping ions as they approach the throat and move through the expander, consistent with the Egedal lost ion treatment [5].

### 8.3 Electron current balance and Eq. 70 density

Ion loss establishes the electron current that must escape for ambipolar current balance. For singly charged D and T,

$$I_i = e \sum_s Z_s \Gamma_s, \quad I_e = I_i. \quad (8.6)$$

The electron barrier energy is  $E_w = e[\phi(0) - \phi_w]$ . Egedal Eq. 68 gives a high barrier approximation to the collisional electron tail loss [5]. The code instead evaluates the corresponding exact tail refilling integral and numerically inverts it to obtain the barrier that matches the ion current. The Eq. 68 approximation remains a diagnostic reference. The escaping electron energy is treated similarly, with Egedal Eq. 69 retained as the high barrier comparison. In the exact tail refilling implementation, the normalized particle loss kernel is the exponential integral  $E_1(y)$  with  $y = E_w/T_e$ . Its large barrier limit approaches  $e^{-y}/y$ , which reproduces the asymptotic dependence used by Egedal Eq. 68.

The distinction between the parent Maxwellian density and the confined electron density is important during this inversion. The current balance can determine a barrier for a trial parent density, while the full Eq. 70 profile determines the parent normalization required by quasineutrality. The current and density closures are therefore reconciled rather than solved as two unrelated scalar problems.

Using the positive electrostatic energy drop

$$\Delta E_\phi(z) = e[\phi(0) - \phi(z)], \quad (8.7)$$

Eq. 70 is evaluated in the equivalent incomplete-gamma form

$$n_e(z) = n_0 \exp \left[ -\frac{\Delta E_\phi(z)}{T_e} \right] P \left( \frac{3}{2}, \frac{E_w - \Delta E_\phi(z)}{T_e} \right). \quad (8.8)$$

Here  $n_0$  is the parent Maxwellian density, which differs from the confined midplane electron density when the velocity tail is truncated by the wall barrier.

### 8.4 Electrostatic accessibility and local reconstruction

The electrostatic potential changes local ion accessibility even though the active Eq. 61 loss boundary remains fixed. In the implementation sign convention, Egedal Eq. 71 becomes [5]

$$\Lambda_{\text{TP}}(U) = \frac{1}{R_M} \left( 1 + \frac{\Delta E_{\phi, \text{throat}}}{U} \right). \quad (8.9)$$

When the ordinary confined interval remains open, Egedal Eq. 72 maps that electrostatically compressed interval back to the magnetic reference coordinate

$$\Lambda^* = 1 - \frac{1 - 1/R_M}{1 - \Lambda_{\text{TP}}} (1 - \Lambda). \quad (8.10)$$

The invariant distribution is evaluated through this mapping and transformed to local speed and signed pitch while conserving total energy and magnetic moment.

Eqs. (8.9) and (8.10) do not replace the fixed loss boundary in Eq. (8.1). They control electrostatic accessibility and local reconstruction. The selected low energy feedback model can also modify the first modal scattering eigenvalue where the accessible interval closes, but the Eq. 61 boundary itself remains magnetic and fixed.

## 8.5 Axial quasineutrality solve

For the NBI supported stationary closure, the final positive-charge density is the solved fast D plus fast T charge. Startup background profiles are not added to the final physical quasineutral state. More general internal electrostatic routines can accept an explicit background term for other closures.

The central condition is

$$n_e(z) = n_+(z) = \sum_s Z_s [n_{s,\text{conf}}(z) + n_{s,63}(z)] \quad (8.11)$$

for the active stationary D and T closure. Each axial cell is solved for a bounded  $\Delta E_\phi(z)$  between the midplane reference and wall barrier. The previous profile is used as a continuation seed. If it does not give a valid local bracket, the code searches the allowed interval.

The exact magnetic midplane and exact left and right throat states are solved independently rather than inferred from nearby cell centers. This prevents grid placement from changing the scalar reference state used by current balance and density normalization.

After a potential profile is obtained, local ion distributions and the Eq. 63 populations are rebuilt and reconciled with Eq. 70 until the electrostatic state satisfies the configured convergence criteria. The final electrostatic result retains the cell profile, exact anchor nodes, parent electron density, species resolved ion densities, accessibility diagnostics, and local velocity distributions.

The major closure identities include Eq. 61 loss consistency, Eq. 63 differential flux matching, ion-electron current balance, exact midplane and exact throat quasineutrality, bounded Eq. 70 roots, volume integrated density consistency, and convergence of the Eq. 63 to Eq. 70 reconciliation.

## 9 Coupled D and T Operating Point and Electron Temperature Closure

A complete neutral-beam-supported operating point requires the beam, kinetic, electrostatic, expander, current, and electron energy states to be mutually consistent. Beam attenuation depends on the plasma it creates. The D and T kinetic distributions depend on the beam source, collision state, and ambipolar potential. Ion losses determine the electron barrier and terminal current. The electron temperature depends on collisional heating and end loss power. The model therefore contains several nested fixed points rather than one independent solve for each quantity.

At a fixed electron temperature, the source model iterates the beam stopping target and the coupled D and T kinetic state until the beam source and solved density agree. Each fast species retains its own source, speed grid, Eq. 59 amplitudes, collision response, charge-exchange sink, and ion loss spectrum, while both species share the magnetic geometry, electron state, and Eq. 70 potential. The converged candidate is then rebuilt so final acceptance depends on the stationary state itself rather than only on the change between consecutive iterates. species resolved ionization, charge-exchange gain and removal, and terminal losses are checked through the stationary particle balance.

The throat current from the central kinetic solve is then coupled to the source connected expander solution so the final current and wall power state are based on particles that actually reach material. Once that fixed temperature state is closed, the represented electron energy residual is evaluated from fast ion collisional heating, external electron heating, electron wall losses, and electrostatic acceleration of escaping ions. In self-consistent mode, every trial  $T_e$  requires this complete inner operating point. The selected temperature is finally rebuilt from a clean state so the reported  $T_e$  belongs to the same qualified plasma state as the beam and neutron source.

## 9.1 Coupled D and T kinetic closure

Each fast species retains its own speed grid, source, Eq. 59 amplitudes, collision response, charge-exchange sink, magnetic loss spectrum, Eq. 63 population, and local distribution. The species share the magnetic basis structure, electron state, wall barrier, and Eq. 70 potential.

For the active NBI supported stationary closure,

$$n_e(z) = n_D(z) + n_T(z) \quad (9.1)$$

because the final maintained physical ion state consists of the solved singly charged fast D and fast T populations. The startup seed remains only an initialization reference.

The scalar kinetic fixed point updates the fast species densities, electron midplane and collision densities, fast self collision operators, reduced fast cross species Rosenbluth fields, shared wall barrier, and Eq. 70 profile until the joint state converges.

## 9.2 Beam density fixed point and stationary particle balance

Beam stopping depends on the plasma that the beam itself helps create. At fixed  $T_e$ , the outer iteration therefore follows

$$\text{target density} \rightarrow \text{beam attenuation} \rightarrow \text{fast source} \rightarrow \text{coupled kinetic solve} \rightarrow \text{updated density}$$

The solved electron density becomes the heavy particle stopping target for the next iteration. Convergence checks include pointwise and volume weighted density changes, species density changes, total fast birth rate, deposited power, axial birth profile, and energy-component deposition.

Passing these change metrics does not by itself define the final state. The candidate density is used to rebuild the beam source and the kinetic calculation is repeated. The NBI supported stationary closure also checks the species particle balances

$$\dot{N}_{\text{ion},s} + \dot{N}_{\text{cx,gain},s} - \dot{N}_{\text{cx,sink},s} = \dot{N}_{\text{terminal},s}. \quad (9.2)$$

Ionization is a net source. Cross-isotope charge exchange transfers inventory between species, while same isotope charge exchange redistributes velocity space population without changing the total isotope inventory.

## 9.3 Terminal current and represented electron energy balance

The central kinetic solve first provides a throat level current state. The source connected expander calculation then determines which lost ions actually reach material. Those terminal rates are fed back into the shared current balance. If the updated current state changes the plasma enough to invalidate the beam density closure, the beam and kinetic state are rebuilt.

Once the fixed temperature beam, kinetic, expander, and terminal current state is evaluable, the represented electron energy balance is formed. Its heating is

$$P_{e,\text{heat}} = P_{\text{fast} \rightarrow e} + P_{e,\text{external}} \quad (9.3)$$

and its represented losses are

$$P_{e,\text{loss}} = P_{e,\text{wall}} + P_{i,\text{accel}}. \quad (9.4)$$

The scalar residual is  $R_E = P_{e,\text{heat}} - P_{e,\text{loss}}$ . This use of collisional electron heating and end loss energy balance follows the mirror energy-balance logic used by Egedal et al. and related mirror modeling [5, 7].

The model labels this quantity as a represented confined plasma electron energy balance. It does not claim a complete global plasma power balance. Effects such as turbulence, impurity radiation, and other unrepresented device level power channels are outside this closure unless introduced through an explicit represented term.

## 9.4 Fixed and self-consistent electron temperature

In fixed temperature mode, the complete operating point is evaluated at the configured  $T_e$  and  $R_E$  is retained as a diagnostic.

In self-consistent mode, each trial temperature requires a complete physical operating point evaluation. A temperature residual is admitted to the scalar root solver only when the beam density fixed point, kinetic state, terminal current, expander potential, and represented energy terms are evaluable. Invalid inner states are rejected rather than interpreted as values of  $R_E(T_e)$ .

The configured temperature interval is scanned for a valid sign changing bracket. Nearby valid trials can reuse a compatible warm state. After the scalar solve converges, the selected temperature is rebuilt from a clean state. Final acceptance requires the clean residual to satisfy the configured power tolerance, remain consistent with the trial residual, and pass the required kinetic, current, expander, beam density, and energy qualifications. The final  $T_e$  is therefore the temperature of a qualified coupled state rather than only the zero of a standalone scalar function.

The nested operating point hierarchy can be summarized as follows.

1. Solve the coupled D and T kinetic and electrostatic state for one beam target and one electron temperature
2. Update beam attenuation from the solved target state and close the beam density fixed point
3. Rebuild the candidate beam and kinetic state for final beam consistency
4. Map the throat losses into the expanders and determine the terminal ion current
5. Reconcile the terminal current with the shared electron barrier and verify postclosure beam consistency
6. Evaluate the represented electron energy residual
7. Repeat the entire fixed temperature state for additional trial temperatures when self-consistent  $T_e$  is requested
8. Rebuild the selected temperature from a clean state for final qualification

This hierarchy is computationally expensive because each scalar temperature trial contains several kinetic and transport like fixed points. The explicit separation is nevertheless useful because each loop has its own physical meaning and independent convergence evidence.

## 10 Expander and full device Fast Ion Populations

The central FBIS solution is defined between the magnetic throats, but particles that cross those throats continue through the expanding magnetic field before reflecting or striking material. Those source connected particles contribute to the full device fast ion inventory, terminal current, wall power, and fusion outside the central confined region. The expander model therefore converts the directed Eq. 63 throat flux into a local population over the physical device.

Each throat source state is represented by conserved total energy and magnetic moment. The Eq. 63 pitch dependence is sampled with equal probability quantiles so every invariant speed bin preserves its differential Eq. 61 rate. The trajectory is then followed through the local magnetic field and electrostatic potential. Accessibility is determined from the parallel kinetic energy. A source state can remain material connected, reflect and return toward the throat, or encounter source disconnected local well topology. Reflected particles remain part of the source connected population, while disconnected local wells are identified diagnostically but are not assigned a steady state occupancy.

Particle rate is converted into local inventory through the finite transit time along the expanding flux tube path. Material geometry and the prescribed radial profile determine what fraction of that source remains accessible as the flux tube expands. The expander Eq. 70 potential is solved separately on each side from quasineutrality using the same electron parent state and wall barrier as the central solution. The final full device D and T distributions combine the confined population, the central directed loss population, and the source connected expander inventory. These distributions then become the authoritative reactants for fusion.

### 10.1 Directed throat source and adiabatic trajectory

The expander starts from the left and right Eq. 63 fluxes rather than from the entire confined distribution. Within one invariant speed bin it is convenient to use

$$x = 1 - R_M \Lambda, \quad 0 \leq x \leq 1. \quad (10.1)$$

At fixed  $U$ , the Eq. 63 throat flux dependence in this coordinate is proportional to

$$p(x) \propto \exp\left(-\frac{U}{T_L}x\right). \quad (10.2)$$

The implementation represents this continuous pitch dependence with equal probability quantiles rather than ordinary  $\Lambda$  cell midpoints. Each invariant speed bin is divided among the configured number of pitch quantiles, and the quantile weights sum to the physical one end Eq. 61 rate in that speed bin. This preserves the differential loss spectrum while giving the trajectory solver a finite set of source states distributed according to the Eq. 63 pitch dependence.

For each source state the code retains  $U$  and  $\mu B_0 = \Lambda U$ . Using the same positive potential energy convention as Section 8, the local parallel kinetic energy is

$$K_{\parallel}(z) = U + Z \Delta E_{\phi}(z) - \mu B_0 \tilde{B}(z). \quad (10.3)$$

A trajectory is accessible only while  $K_{\parallel} \geq 0$  continuously from the throat. A particle cannot cross an inaccessible region simply because a second positive-energy interval exists farther downstream.

Source states are classified as material connected, reflected, source disconnected local well topology, or unresolved. Reflected particles remain source connected and their return branch is represented with the opposite signed pitch. A disconnected local well is identified diagnostically but is excluded from the final fast ion population because the directed throat source does not determine its steady state occupancy.

### 10.2 Finite transit inventory and material accessibility

A directed rate is converted to a local inventory using transit time. For one path segment,

$$N_{\text{segment}} = \dot{N} \Delta t, \quad \Delta t = \frac{\sqrt{2m} \Delta s}{\sqrt{K_{\parallel,0}} + \sqrt{K_{\parallel,1}}}. \quad (10.4)$$

Only the accessible fraction of a segment is counted when a trajectory turns inside it. Local speed and pitch are reconstructed from the conserved invariants, and the inventory is deposited onto the local  $(v, \xi)$  grid. Reflected source states contribute both outward and return residence time.

The nominal flux tube expands as  $a(z) = a_0/\sqrt{\widetilde{B}}$ . The reduced path length between two neighboring axial locations includes that radial expansion,

$$\Delta s = \sqrt{(\Delta z)^2 + (\Delta a)^2}. \quad (10.5)$$

Using  $\Delta s$  rather than only  $\Delta z$  makes the finite transit inventory follow the expanding flux tube path used by the reduced geometry.

### 10.3 Radial material accessibility

The kinetic FBIS solution is axial and does not solve a self-consistent radial kinetic distribution. The expander therefore uses a prescribed normalized radial profile to route the source connected population through the available flux tube. The normalized coordinate is

$$\rho = \frac{r}{a(z)}. \quad (10.6)$$

At each path location, the surrounding material geometry determines a largest accessible normalized radius  $\rho_{\text{lim}}(z)$ . As the nominal flux tube expands or intersects material,  $\rho_{\text{lim}}$  can decrease. Integrating the prescribed radial profile over the accessible interval determines the surviving source connected fraction. Population removed by this radial accessibility calculation is assigned to the corresponding material surface, while the remaining fraction continues outward. This is a reduced material routing closure and does not replace the one-dimensional FBIS solution with a radially resolved kinetic solve.

### 10.4 Expander Eq. 70 potential and terminal state

Each expander uses the central throat value  $\Delta E_{\phi, \text{throat}}$  as its inner electrostatic boundary and the shared wall barrier  $E_w$  as its outer boundary. For every trial potential profile, the source connected ion trajectories are remapped and their local positive charge density is recomputed. The electron density is evaluated with the same parent density  $n_0$ , electron temperature, wall barrier, and Eq. 70 truncated Maxwellian model used by the central solution.

The local quasineutrality residual is represented as

$$R_n(z) = \frac{n_+(z) - n_e(z)}{\max[n_+(z), n_e(z)]}. \quad (10.7)$$

The cell center values of  $\Delta E_\phi(z)$  are varied within the physical interval between the midplane reference and  $E_w$  to reduce these residuals. A linear throat-to-wall potential is used only as the initial guess. After the cell center solve, the potential is reconstructed piecewise between the fixed throat value, the solved interior values, and the fixed wall value. The expander state is accepted when the maximum required quasineutrality residual satisfies the configured tolerance.

Material connected particle rates from this converged state determine the terminal ion current. Their change in kinetic energy relative to the midplane invariant determines the electrostatic acceleration power used by the electron energy balance. The source connected trajectory mapping and expander potential solution are implemented primarily in `expander/mapping.py` and `expander/solver.py`.

The final species distribution is assembled as

$$f_{s, \text{full}} = f_{s, \text{conf}} + f_{s, 63} + f_{s, \text{exp}}. \quad (10.8)$$

The confined state is conservatively remapped to the full device axial grid. Eq. 63 supplies the directed lost population inside the central region, and the source connected expander solve supplies the population outside the throats. source disconnected local well occupancy is not included.

Before this state becomes the fusion authority, the expander stage checks particle rate partition, energy partition, transit inventory, local velocity grid coverage, quasineutral potential convergence, terminal current consistency, terminal surface alignment, and the unresolved source connected fraction. These checks preserve the distinction between crossing the magnetic throat and actually reaching material.

The expander population also carries the electrostatic acceleration of material connected ions. The terminal wall power is calculated from the particles that reach material rather than assigning the wall potential energy to every throat crossing ion. This distinction is required by the terminal current and electron energy closures in Section 9.

## 11 Fusion and Neutron Source Calculation

The fusion stage converts the solved full device D and T distributions into physical reaction rates without replacing those non-Maxwellian, anisotropic populations by an effective ion temperature. The local reaction kernel integrates directly over both reactant velocity distributions and the relative gyroangle. The total fusion cross sections are the Bosch and Hale Table IV parametrizations [4], and the code tracks the reaction rate fraction that lies inside and outside their stated center-of-mass fit intervals.

The active reaction set contains the D-T neutron branch and both principal D-D branches. D-D reactions use the one half identical particle factor. The pair kernel retains reaction resolved axial rate density, integrated reaction rate, fusion power, neutron producing rate, charged product power, and reaction weighted reactant kinetic energy removal.

The deterministic neutron calculation uses the same reactant pairs and the same  $\sigma g$  physical reaction weights. Relativistic two-body kinematics maps each moving reactant pair to the allowed laboratory neutron energies following the arbitrary reactant framework of Eriksson et al. [8]. For the deterministic spectrum, the exact isotropic center-of-momentum energy marginal is integrated analytically rather than sampled. The raw neutron rate is then checked directly against the neutron producing fusion rate, while represented and out-of-range rates remain separate. This makes the deterministic neutron spectrum both a source representation and an independent conservation check on the fusion calculation.

### 11.1 Arbitrary distribution fusion kernel

For local reactant distributions  $f_a$  and  $f_b$ ,

$$R_{ab}(z) = C_{ab} \int d^3v_a \int d^3v_b f_a f_b \sigma(E_{\text{cm}}) g, \quad (11.1)$$

where  $g = |\mathbf{v}_a - \mathbf{v}_b|$  and  $C_{ab} = 1$  for distinct populations or  $1/2$  for identical populations. The CM kinetic energy is  $E_{\text{cm}} = \mu_{ab} g^2 / 2$  in the nonrelativistic cross section evaluation.

Because the local distributions are gyrotropic, the absolute gyroangle of either reactant is irrelevant but their relative gyroangle  $\Delta\phi$  changes the relative speed. For local pitches  $\xi_a$  and  $\xi_b$ ,

$$g^2 = v_a^2 + v_b^2 - 2v_a v_b \left[ \xi_a \xi_b + \sqrt{1 - \xi_a^2} \sqrt{1 - \xi_b^2} \cos \Delta\phi \right]. \quad (11.2)$$

The code therefore evaluates the gyrophase dependence with deterministic quadrature over  $\Delta\phi$  rather than constructing a full three-dimensional Cartesian velocity grid for either reactant. Only represented velocity cells are included and large pair spaces are processed in batches. The arbitrary distribution pair integration is implemented in `fusion/reactivity.py` and `fusion/pair_kernel.py`.

The active branches are  $D + T \rightarrow n + \alpha$ ,  $D + D \rightarrow n + {}^3\text{He}$ , and  $D + D \rightarrow p + T$ . The first two contribute to the neutron source. The proton branch contributes to fusion power and D burnup.

Bosch and Hale write the total cross section as [4]

$$\sigma(E) = \frac{S(E)}{E} \exp\left(-\frac{B_G}{\sqrt{E}}\right), \quad (11.3)$$

where the reaction specific astrophysical factor is represented by the Bosch and Hale Padé form

$$S(E) = \frac{A_1 + E[A_2 + E(A_3 + E[A_4 + EA_5])]}{1 + E[B_1 + E(B_2 + E[B_3 + EB_4])]} \quad (11.4)$$

Separate Table IV coefficient sets are used for the D-T neutron branch, the D-D neutron branch, and the D-D proton branch. The code also retains the Table IV CM fit interval for each reaction and records how much reaction weight falls outside that interval rather than silently clipping the reacting energy. The cross section implementation is contained in `fusion/bosch_hale.py` and `fusion/cross_sections.py`.

The local reaction density is converted to integrated reaction rate using the physical flux tube cell volumes. For reaction  $r$ , the axial source profile stores the reaction density, neutron producing rate when applicable, total fusion power, neutron power, charged product power, and reaction-weighted reactant kinetic energy removal. Reaction branches are combined only after their individual rates have been evaluated.

## 11.2 Deterministic neutron spectrum

The deterministic neutron spectrum uses the same reactant pairs, Bosch and Hale cross section, and  $\sigma g$  reaction weight as the fusion calculation. It changes only how the neutron producing reaction rate is distributed in neutron energy.

Neutron energy is computed from the actual reactant velocities with relativistic two-body kinematics following the arbitrary reactant framework described by Eriksson et al. [8]. For a fixed reactant pair and isotropic CM emission, the laboratory neutron energy is linear in the CM emission cosine. The deterministic spectrum therefore integrates the exact isotropic CM energy marginal analytically between its relativistic kinematic endpoints instead of sampling emission directions.

The spectrum is stored as an axial by energy matrix. The calculation separately retains the raw physical neutron rate, the represented rate within the configured energy bounds, and any out-of-range rate. The represented spectrum is not rescaled to the fusion rate after calculation. Instead, the raw neutron rate is independently checked against the neutron producing fusion rate and the axial profiles are checked for the same consistency.

The neutron stage keeps three rate concepts distinct. The raw rate is the complete rate produced by the reactant pair quadrature. The represented rate is the portion contained inside the configured neutron energy interval. Their difference is the out-of-range rate. A material out-of-range contribution is treated as a failed source representation rather than silently discarded.

The D-D and D-T spectra remain separate until the total neutron source is formed. This preserves reaction identity for diagnostics and for the correlated event stage. The neutron stage also compares the raw axial production profile with the fusion stage neutron profile, so agreement of only the integrated total is not sufficient to hide a spatial normalization error.

## 12 Correlated Neutron Events and OpenMC Source Export

The deterministic neutron spectrum provides the physical rate and a low noise energy marginal, but neutron transport also depends on the correlation among birth position, laboratory energy, and laboratory direction.

The correlated event stage therefore samples individual D-D and D-T reactant pairs from the same full device distributions used by the fusion calculation. The physical stratum rates are known before sampling, so finite Monte Carlo events redistribute an already determined reaction rate rather than estimating that rate from the event count.

For each sampled pair, the Bosch and Hale total cross section supplies the scalar reaction importance and the packaged ENDF/B-VIII.1 angular law supplies the normalized center-of-momentum emission probability. Because both reactants can be moving, the pair invariant mass is first converted to the equivalent stationary target deuteron energy used by the evaluated angular representation. D-D uses its tabulated probability density in emission cosine, while D-T uses interpolated Legendre coefficients. The sampled center-of-momentum neutron is then Lorentz transformed to the laboratory frame so neutron energy and direction remain correlated with the sampled reactant velocities.

The event bank also assigns a three-dimensional birth position using the authoritative axial fusion rate and the prescribed radial source profile. Event probabilities sum to one, while the independent physical neutron rate remains in  $\text{s}^{-1}$ . OpenMC export converts this probability-normalized bank into an HDF5 `FileSource` without embedding the physical source rate into the particle weights. After writing, the file is read back and checked for particle fields, joint energy direction preservation, and source bank consistency before it is accepted as the validated transport source.

## 12.1 Reactant Pair and Angular Sampling

Each neutron producing fusion component and positive-rate axial cell defines an event stratum. The requested event count is distributed approximately in proportion to the physical stratum rates, with at least one event assigned to every positive rate stratum.

Within one stratum, reactant speed and pitch cells are sampled from their represented gyrotropic distributions. A uniform absolute gyroangle is assigned to one reactant and the relative gyroangle is selected from the same midpoint quadrature used by the fusion kernel. The sampled pair is weighted by  $q = \sigma(E_{\text{cm}})g$ .

For D-T, the deuteron is the projectile used to interpret the evaluated angular data. For D-D, the reactant labels are randomly exchanged with equal probability so the artificial ordering of two identical deuterons does not bias the emission distribution.

The physical stratum rate is known before event sampling. The sampled  $q$  values only distribute that known rate among finite events. If event  $j$  belongs to stratum  $s$ ,

$$p_j = \frac{R_s}{R_{\text{tot}}} \frac{q_j}{\sum_{k \in s} q_k}, \quad \sum_j p_j = 1. \quad (12.1)$$

Increasing the event count therefore changes sampling resolution, not the calculated physical neutron rate.

The ENDF/B-VIII.1 angular data use a deuteron projectile on a stationary target. The reactants in the source model can both be moving, so the evaluated incident energy cannot be taken directly from either laboratory reactant energy. For the sampled pair, the Lorentz invariant is

$$s = E_{\text{tot}}^2 - |\mathbf{p}_{\text{tot}}c|^2. \quad (12.2)$$

The equivalent laboratory kinetic energy of a deuteron incident on a stationary reaction target is then

$$K_{d,\text{eq}} = \frac{s - (m_d c^2 + m_{\text{target}} c^2)^2}{2m_{\text{target}} c^2}. \quad (12.3)$$

Here  $m_{\text{target}}$  is the deuteron target mass for D-D and the triton target mass for D-T. This mapping

preserves the pair invariant mass while expressing the moving-reactant event in the stationary target energy coordinate used by the evaluated angular law.

For a given  $K_{d,\text{eq}}$ , the evaluated data define a normalized CM cosine probability density

$$\int_{-1}^1 p(\mu \mid K_{d,\text{eq}}) d\mu = 1, \quad \mu = \cos \theta_{\text{cm}}. \quad (12.4)$$

With uniform CM azimuth, the corresponding differential cross section can be written as

$$\frac{d\sigma}{d\Omega} = \frac{\sigma_{\text{BH}}(E_{\text{cm}})}{2\pi} p(\mu \mid K_{d,\text{eq}}). \quad (12.5)$$

The total reaction rate therefore remains controlled by the Bosch and Hale cross section, while the ENDF law determines the CM angular distribution of that rate.

The D-D neutron data are represented as a tabulated probability density in  $\mu$ . The sampler uses piecewise-linear inversion at the evaluated energy knots together with the prescribed linear energy interpolation. The D-T data are represented by Legendre coefficients. Those coefficients are interpolated in incident energy and the resulting angular probability is sampled by rejection sampling. These distinct representations are preserved rather than converting both reactions to one approximate angular form.

Before use, the packaged angular datasets are checked for the required reaction identity, CM reference frame, supported representation, nonnegative probability, normalization, and incident energy coverage. Each sampled pair must lie inside the evaluated incident energy interval. A numerically negligible out-of-range sampled importance can be retained only as a zero weight audit event, while a material out-of-range contribution invalidates the event construction. The runtime loader validates these physical data properties. The separate archive validator additionally contains the packaged provenance and hash checks. The incident energy mapping and angular samplers are implemented under `nuclear_data/endl_b_viii1/` and `neutrons/events/`.

## 12.2 Relativistic event kinematics and spatial sampling

The sampled CM neutron direction is defined relative to the deuteron projectile direction in the CM frame. The final state momentum magnitude is set by two-body energy and momentum conservation, then the neutron four-momentum is Lorentz transformed back into the laboratory frame. The same event therefore retains the physical correlation between neutron energy and direction [8].

The event generator also reconstructs the residual product and verifies its mass shell. Events retain the sampled reactant velocities, CM energy, equivalent evaluated incident energy, CM emission cosine, reaction identity, and axial cell so the kinematic source can be audited after generation.

The event bank therefore contains more information than the OpenMC particle source requires. The detailed state is retained so angular law behavior, incident energy coverage, reactant pair sampling, and lab frame transformations can be checked without attempting to infer them from the final HDF5 particle bank.

The axial birth distribution is authoritative from the fusion calculation. A prescribed normalized radial profile distributes that rate across the local cross section without changing the axial fusion rate. The current profile uses

$$g_r(\rho) = 2(1 - \rho^2), \quad p_r(\rho) = 4\rho(1 - \rho^2), \quad \rho = \frac{r}{a(z)}. \quad (12.6)$$

The second expression includes the cylindrical area measure. Axial position is sampled within the selected source cell and azimuth is uniform.

### 12.3 Event normalization and OpenMC export

The `CorrelatedNeutronEventBank` stores position, direction, neutron energy, normalized probability, reaction and component labels, axial cell, CM quantities, and both reactant velocities. It also stores the independent physical total rate  $R_{\text{tot}}$  in  $\text{s}^{-1}$ . Event  $i$  therefore represents the physical rate  $R_i = p_i R_{\text{tot}}$ .

For OpenMC export, model positions are converted from meters to centimeters and energies from joules to electron volts. Only positive probability events are written. If  $N_+$  events are exported, OpenMC source weights are

$$w_i = N_+ p_i, \quad \sum_i w_i = N_+, \quad \langle w \rangle = 1. \quad (12.7)$$

The physical neutron rate is not embedded in these particle weights. It remains a separate metadata quantity used to normalize OpenMC results. The exported `FileSource` therefore describes the source probability distribution while the source model supplies its absolute strength.

The validated source bundle contains the OpenMC HDF5 source bank, a source file validity record, and sampling metadata. A correlated event NPZ sidecar can retain the more detailed reactant and angular audit state that does not fit in the standard OpenMC particle record.

The sampling metadata records the physical source rate, normalization convention, requested and represented event counts, reaction composition, random seed, and source construction diagnostics. The validity record is deliberately separate because it describes measured properties of the written file rather than the intended in-memory event state.

After writing, the HDF5 structure and required `source_bank` fields are checked and the file is then read back through OpenMC itself. The measured roundtrip comparison includes particle count, position, direction, energy, source time, particle weight, and neutron identity. The exporter also compares the energy direction covariance before and after serialization and directly compares the complete joint energy and direction rows. These joint checks are stronger than comparing separate one-dimensional energy and angular distributions because they verify that the laboratory correlation produced by the relativistic event calculation survives conversion to the OpenMC source format.

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